

Calculus II Project: Taylor Series of the Function $f(x) = \tan x$

Criteria	Exceptional (4)	Proficient (3)	Developing (2)	Beginning (1)
Derivatives/ Calculation	Correctly computes derivatives up to 5th derivative or higher.	Computes up to 3rd or 4th derivative correctly.	Makes errors in higher-order derivatives (> 3) but understands the method.	Significant errors in first/second derivatives.
Series Structure	Correctly identifies and utilizes that $\tan(x)$ is odd, setting even terms to 0.	Recognizes $\tan(x)$ is odd but struggles with subsequent calculations.	Ignores the odd property, resulting in unnecessary calculations.	Does not understand the structure of a Taylor series.
Coefficient Evaluation	Correctly evaluates coefficients (e.g., 1, $\frac{1}{3}$, $\frac{2}{15}$) and simplifies fully.	Evaluates coefficients with minor errors or leaves in factorial form (e.g., $\frac{2}{5!}$).	Evaluates 1-2 terms correctly but misses or wrongly calculates higher order terms.	Incorrect coefficients for all terms.
Expansion & Form	Writes the polynomial correctly with correct powers and factorial denominators.	Writes the polynomial with minor errors in powers of x .	Inverts derivative formula (incorrect factorial usage).	Incorrect formula used.

Convergence	Correctly identifies the Radius of Convergence as $R = \pi/2$.	Identifies $R = \pi/2$ but cannot fully explain why.	Understands convergence is limited but gives wrong value.	No understanding of convergence.
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